Matthew Lin

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Education

Rensselaer Polytechnic Institute Troy, NY
B.S. in Computer Science and Information & Web Technologies *Expected Dec. 2028*

- **GPA:** 3.70/4.00
- **Coursework:** Data Structures, Discrete Mathematics and Computing, Computer Architecture and Operating Systems, Linear Algebra

Experience

Software Development Internship Jul. 2026 – Aug. 2026
NYC DOE – Division of Instructional and Information Technology *New York, NY*

- Fixed document corruption during overlapping uploads by consolidating writes into a single database transaction; tested with sample uploads and obtained supervising engineer approval.
- Refactored a large legacy ASP.NET Core document-processing function into smaller, reusable methods for individual processing steps.
- Consolidated registration database writes into a single transaction to keep related updates consistent.

Web Development & Project Coordination May 2025 – Jun. 2025
Nebraska Wesleyan University – WISE Program

- Led a six-person team to deliver a working promotional website prototype for a university program, submitting the completed prototype for review.
- Implemented site navigation, dropdown menus, and frontend styling using HTML, CSS, and JavaScript.
- Maintained a shared task list, divided frontend and backend work, and coordinated code contributions with Git to support asynchronous collaboration.

Software Development Internship Mar. 2025 – Aug. 2025
Bridge to College *Remote, Part-time*

- Developed player movement and environmental interactions in Unity and C# for a completed HBCU campus-exploration prototype, including building entry and school-information panels.
- Designed the application's menu and campus environment in Figma to guide prototype development.

Projects

GoTo: Transit Desert Analysis | *Python, HTML* Overall 1st Place

- Collected and cleaned population-density, income, and bus-stop data in Python, standardizing dataset formats and filtering irrelevant records for regional transit-access analysis.
- Collaborated on a four-person team's HV Open Data Hackathon demo displaying a regional transit-access score map, proposed bus routes, and route extensions.

Mess Environment Sensor System | *ESP32, ESP-IDF* Sep. 2026 – Present

- Assembled ESP32 hardware and flashed firmware with ESP-IDF on a three-person team for a Western Digital-sponsored indoor-environment monitoring project.
- Verified BME280 and BH1750 sensors for temperature, humidity, pressure, and light measurements.

Better EHR | *React, Node.js, Firebase, WebSockets* HackRPI 2025

- Collaborated on a four-person team to build a full-stack electronic health record system with patient record creation and retrieval, PDF export, and real-time physician messaging using React, Express, Firebase Firestore, and WebSockets.
- Integrated AssemblyAI streaming speech-to-text with Gemini 2.0 Flash to transcribe doctor-patient conversations and generate structured clinical summaries from visit transcripts.

Technical Skills

Languages: C, C++, C#, Java, Python, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: ASP.NET Core, React, Express.js, Bootstrap, Unity
Databases & Backend: Firebase Firestore, Node.js, REST APIs, WebSockets
Tools & Platforms: Git, Figma, ESP32, ESP-IDF
APIs & Technologies: Gemini API, AssemblyAI API